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"JOBGUARD AI"

IMPLEMENTATION PHASE - 1

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1. Introduction

JobGuard AI is a project developed to assist in identifying potentially fraudulent job postings and suspicious URLs. The system combines job-posting text analysis, URL-feature analysis and supporting detection logic to help users assess whether an opportunity or web address may be unsafe. This report presents the measured performance of the supervised machine-learning pipelines evaluated for the project.

The evaluation covers two separate classification tasks. The EMSCAD dataset is used for job-posting fraud detection, where text from job advertisements is converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF) and classified using Logistic Regression. The PhiUSIIL dataset is used for phishing URL detection, where 18 numerical URL features are evaluated with HistGradientBoosting, XGBoost and Logistic Regression in the experiments described in this report.

The datasets are not merged into one trained classifier. EMSCAD contains job-advertisement text and its target indicates whether a job posting is fraudulent. PhiUSIIL contains URL-related features and its target indicates whether a URL is phishing. Since the tasks, inputs and labels differ, their scores are reported separately. Any arithmetic mean shown for context is explicitly identified as a descriptive summary, not as the performance of a combined model.

The report is organised around measured results, graphical interpretation and discussion. The presentation follows the results-oriented approach used in Chapter 4 of the supplied Kidney.pdf—tables, performance plots and ROC analysis—and the interpretation-led discussion approach in its Chapter 5. Only the presentation structure is adopted; all methods, figures and findings in this report relate to JobGuard AI.

2. Evaluation Setup

2.1 Datasets

Dataset	Records	Input used	Positive class
EMSCAD	17,880	Job-posting text fields: title, company profile, description, requirements and benefits	Fraudulent job posting (1)
PhiUSIIL	235,795	18 numerical URL-derived features	Phishing URL (mapped to 1 for evaluation)

EMSCAD contains 17,880 annotated job advertisements. The text pipeline combines the five listed text fields into one document per record. TF-IDF is fitted on the training text and then applied to the test text. The measured multi-split pipeline uses Logistic Regression.

PhiUSIIL contains 235,795 URL records. The evaluation uses the following 18 numerical features: URLLength, DomainLength, IsDomainIP, TLDLength, NoOfSubDomain, HasObfuscation, NoOfObfuscatedChar, ObfuscationRatio, NoOfLettersInURL, LetterRatioInURL, NoOfDegitsInURL, DegitRatioInURL, NoOfEqualsInURL, NoOfQMarkInURL, NoOfAmpersandInURL, NoOfOtherSpecialCharsInURL, SpacialCharRatioInURL and IsHTTPS.

The source PhiUSIIL label convention is 0 for phishing and 1 for legitimate URLs. For the reported binary evaluation, the target is mapped so that phishing is the positive class (1). This mapping is important when interpreting precision, recall, false positives and false negatives.

2.2 Train–test split design

The multi-split experiment evaluates 90:10, 80:20, 70:30, 60:40 and 50:50 train–test proportions. The first number is the percentage of records assigned to training; the second number is the percentage assigned to testing. For example, a 70:30 split means 70% of the records are used to fit the model and 30% are held out for evaluation. The split counts below are the counts reported by the evaluation runs.

Split	Training share	Testing share	EMSCAD train / test	PhiUSIIL train / test
90:10	90%	10%	16,092 / 1,788	212,215 / 23,580
80:20	80%	20%	14,304 / 3,576	188,636 / 47,159
70:30	70%	30%	12,516 / 5,364	165,056 / 70,739
60:40	60%	40%	10,728 / 7,152	141,477 / 94,318
50:50	50%	50%	8,940 / 8,940	117,897 / 117,898

The 70:30 evaluation uses a stratified split with random_state=42 in the supplied evaluation script. The text vectorizer is fitted on training data only. The final three-pipeline experiment is a separate 70:30 run and is reported separately from the earlier five-split HistGradientBoosting experiment.

2.3 Evaluated pipelines

Experiment	Dataset	Feature representation	Classifier
Five-split evaluation	EMSCAD	TF-IDF from five job text fields	Logistic Regression
Five-split evaluation	PhiUSIIL	18 numerical URL features	HistGradientBoostingClassifier
Final 70:30	EMSCAD	TF-IDF from five job	Logistic Regression

comparison		text fields	
Final 70:30 comparison	PhiUSIIL	18 numerical URL features	XGBoost (XGBClassifier)
Final 70:30 comparison	PhiUSIIL	18 numerical URL features	Logistic Regression

The HistGradientBoosting results belong to the five-split experiment. XGBoost and the final URL-feature Logistic Regression results belong to the later three-pipeline 70:30 experiment. They are not substituted into the five-split table because they were not evaluated across all five proportions in the supplied final run.

3. Performance Metrics and Definitions

The models are evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC. The first four measures are calculated from class predictions at the selected decision threshold. ROC-AUC is calculated from prediction scores across possible thresholds and therefore provides a threshold-independent measure of ranking discrimination.

Metric	Formula	Interpretation
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Proportion of all test records classified correctly.
Precision	TP / (TP + FP)	Among records predicted as positive, the proportion that are truly positive.
Recall (Sensitivity)	TP / (TP + FN)	Among actual positive records, the proportion detected.
F1-score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall.
ROC-AUC	Area under the ROC curve	Measures how well model scores rank positive records above negative records across thresholds.

3.1 Confusion-matrix terms

Term	Meaning in EMSCAD	Meaning in PhiUSIIL
TP	Fraudulent job posting correctly flagged.	Phishing URL correctly flagged.

TN	Legitimate job posting correctly accepted.	Legitimate URL correctly accepted.
FP	Legitimate job posting incorrectly flagged as fraudulent.	Legitimate URL incorrectly flagged as phishing.
FN	Fraudulent job posting missed and classified as legitimate.	Phishing URL missed and classified as legitimate.

For a fraud-detection system, false negatives deserve particular attention because they represent fraudulent postings or phishing URLs that are not detected. False positives also matter because they can cause legitimate postings or URLs to be flagged unnecessarily.

3.2 ROC curve interpretation

A Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) as the classification threshold changes. TPR is equivalent to recall, while FPR is FP/(FP+TN). A curve that approaches the upper-left corner indicates that the classifier can achieve high sensitivity while maintaining a low false-positive rate. The ROC-AUC value summarises the area under this curve; 0.50 corresponds to random ranking and 1.00 corresponds to perfect ranking.

4. Measured Results

4.1 EMSCAD — TF-IDF and Logistic Regression

Table 1 presents the measured results for the EMSCAD text-classification pipeline across five train–test proportions. Accuracy remains between 96.92% and 97.45%, while precision is 100.00% in each reported split. Recall is lower, ranging from 36.49% to 47.31%. This difference means that the model's positive predictions are highly precise at the selected threshold, but it does not identify every fraudulent posting in the test data.

Split	Train n	Test n	Accuracy	Precision	Recall	F1-score	ROC-AUC
90:10	16,092	1,788	97.43%	100.00%	47.13%	64.06%	99.13%
80:20	14,304	3,576	97.40%	100.00%	46.24%	63.24%	98.51%
70:30	12,516	5,364	97.45%	100.00%	47.31%	64.23%	98.36%
60:40	10,728	7,152	97.27%	100.00%	43.64%	60.76%	98.35%
50:50	8,940	8,940	96.92%	100.00%	36.49%	53.47%	98.13%

4.2 PhiUSIIL — URL features and HistGradientBoosting

Table 2 presents the five-split results for the 18-feature PhiUSIIL pipeline using HistGradientBoostingClassifier. The scores remain within a narrow range: accuracy is approximately 99.72%–99.76%, recall is approximately 99.41%–99.50%, and ROC-AUC is approximately 99.85%–99.89%. These values describe performance on the evaluated partitions; they do not by themselves establish performance on newly collected URLs.

Split	Train n	Test n	Accuracy	Precision	Recall	F1-score	ROC-AUC
90:10	212,215	23,580	99.75%	99.91%	99.50%	99.71%	99.87%
80:20	188,636	47,159	99.76%	99.94%	99.49%	99.71%	99.87%
70:30	165,056	70,739	99.75%	99.95%	99.47%	99.71%	99.89%
60:40	141,477	94,318	99.72%	99.94%	99.41%	99.68%	99.86%
50:50	117,897	117,898	99.74%	99.95%	99.44%	99.69%	99.85%

4.3 Descriptive macro-average across the two multi-split pipelines

Table 3 retains the arithmetic mean values calculated from the EMSCAD and PhiUSIIL HistGradientBoosting results. This table is included only as a compact descriptive summary. It is not a trained model, does not represent one shared test set, and should not be interpreted as a unified JobGuard classifier.

Split	Summed train n*	Summed test n*	Mean accuracy	Mean precision	Mean recall	Mean F1	Mean ROC-AUC
90:10	228,307	25,368	98.59%	99.96%	73.32%	81.88%	99.50%
80:20	202,940	50,735	98.58%	99.97%	72.87%	81.48%	99.19%
70:30	177,572	76,103	98.60%	99.98%	73.39%	81.97%	99.12%
60:40	152,205	101,470	98.50%	99.97%	71.53%	80.22%	99.10%
50:50	126,837	126,838	98.33%	99.97%	67.96%	76.58%	98.99%

*The train and test counts shown in the source summary are the arithmetic sum of the two datasets' counts. The metric columns are arithmetic means of the two pipeline scores. This does not imply that the datasets were concatenated or jointly trained.

4.4 Final three-pipeline comparison at 70:30

A later evaluation compared three selected pipelines at a 70:30 split. The results are shown separately in Table 4 because this run includes XGBoost and a URL-feature Logistic Regression model rather than the HistGradientBoosting pipeline used in the multi-split table.

Dataset	Pipeline	Train n	Test n	Accuracy	Precision	Recall	F1-score	ROC-AUC
EMSCAD	TF-IDF + Logistic Regression	12,516	5,364	97.45%	100.00%	47.31%	64.23%	98.36%
PhiUSIIL	18 URL features + XGBoost	165,056	70,739	99.75%	99.98%	99.44%	99.71%	99.87%
PhiUSIIL	18 URL features + Logistic Regression	165,056	70,739	99.77%	99.99%	99.46%	99.73%	99.86%
Descriptive mean only	Arithmetic mean of the three pipeline scores	—	—	98.99%	99.99%	82.07%	87.89%	99.36%

The descriptive mean row is not a fourth classifier. It averages three metric values from two different datasets and two different tasks; it is included only to preserve the supplied comparison summary.

4.5 Confusion-matrix counts for the final 70:30 experiment

The confusion-matrix counts below are reproduced from the final evaluation output. In EMSCAD, the positive class is fraudulent job posting. In PhiUSIIL, the positive class is phishing URL after the source labels are mapped.

Pipeline	TN	FP	FN	TP
EMSCAD — TF-IDF + Logistic Regression	5,104	0	137	123
PhiUSIIL — URL features + XGBoost	40,450	5	169	30,115
PhiUSIIL — URL features + Logistic Regression	40,453	2	164	30,120

The EMSCAD confusion matrix explains why recall is lower despite 100.00% precision: the model produced no false positive predictions in this test run, but it missed 137 fraudulent postings. The PhiUSIIL models produced very few false positives and false negatives relative to the size of their test set.

5. Graphical Analysis

The following figures present the measured results visually. In Figure 1, each split label is a training-to-testing proportion: 90:10 means 90% training and 10% testing; 80:20 means 80% training and 20% testing; 70:30 means 70% training and 30% testing; 60:40 means 60% training and 40% testing; and 50:50 means 50% training and 50% testing.

Figure 1. Accuracy and ROC-AUC across five train–test splits

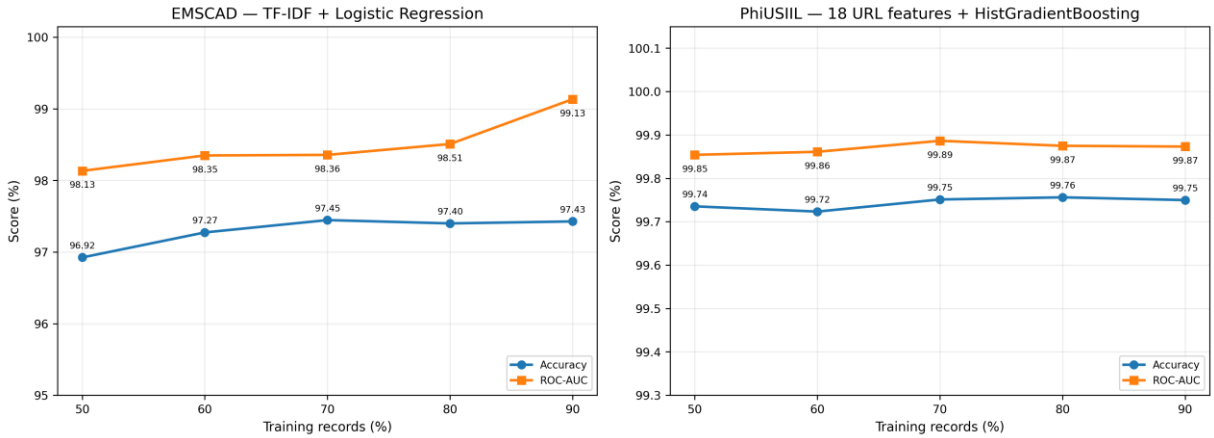


Figure 1. Accuracy and ROC-AUC across the five measured train–test splits.

- Split definition: 90:10 means 90% of records were used for training and 10% for testing; the same interpretation applies to the other split labels.
- EMSCAD panel: accuracy stays close to 97%, while ROC-AUC ranges from approximately 98.13% to 99.13%.
- PhiUSIIL panel: accuracy and ROC-AUC remain close to 100% across all five train–test proportions.
- Interpretation: the small changes indicate similar scores on these particular partitions, not repeated-run stability. Recall is not plotted and remains the main weakness of the EMSCAD threshold-based predictions.

Figure 2. Five-metric comparison of the final 70:30 pipelines

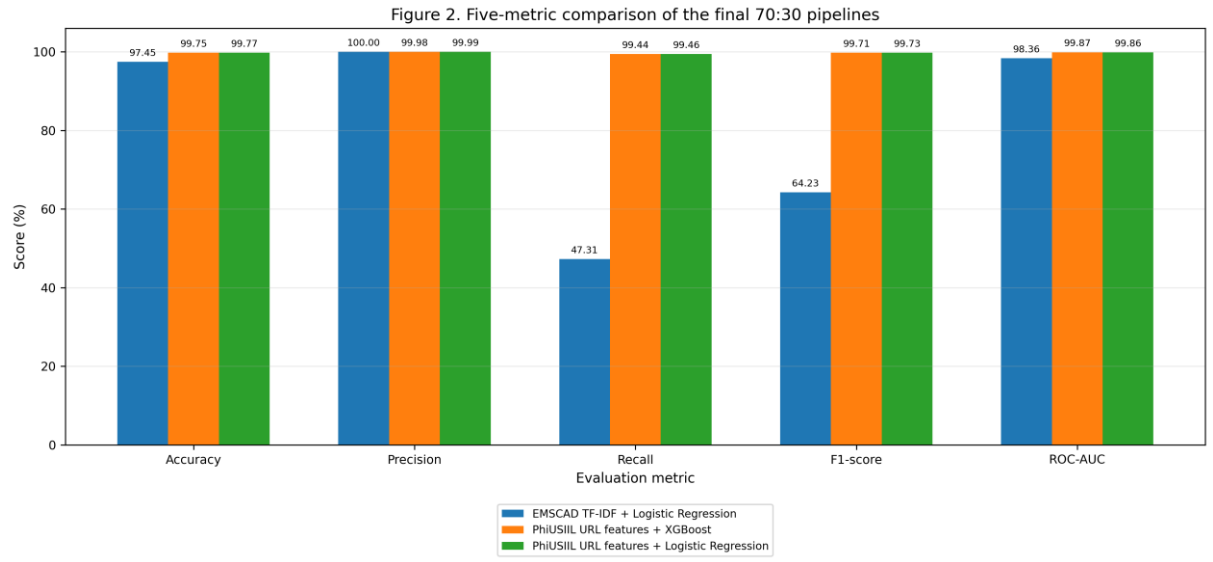


Figure 2. Grouped bar chart comparing five metrics for the three final 70:30 pipelines.

• Metric definition: accuracy is the proportion of all records classified correctly; precision is the proportion of predicted positives that are truly positive.
• Recall definition: recall is the proportion of actual positive records detected. F1-score is the harmonic mean of precision and recall.
• ROC-AUC definition: ROC-AUC measures ranking discrimination across possible decision thresholds, rather than at only one threshold.
• Result: both PhiUSIIL pipelines score close to 100% on all five measures. EMSCAD has high accuracy and precision but lower recall and F1-score.
• Comparison note: the datasets represent different tasks and should not be treated as a direct head-to-head model contest.

Figure 3. Precision–recall profile of the final 70:30 pipelines

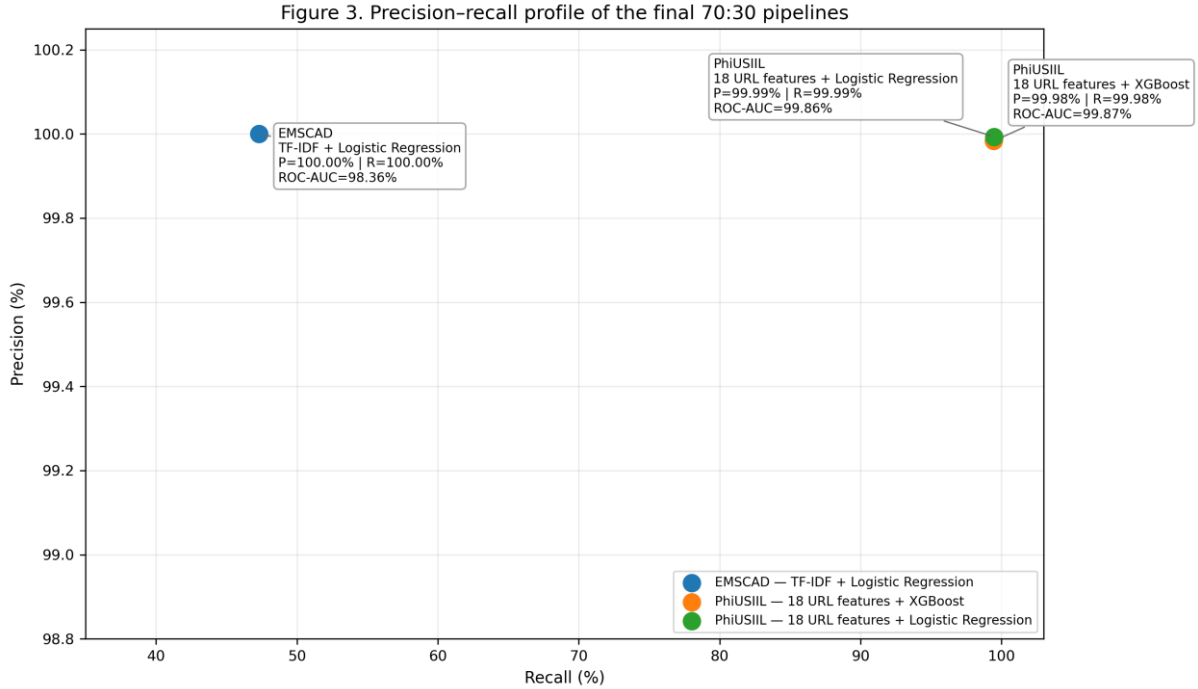


Figure 3. Precision–recall scatter plot with ROC-AUC annotations for the final 70:30 pipelines.

• Horizontal axis: recall, or the proportion of actual fraudulent postings or phishing URLs detected.
• Vertical axis: precision, or the proportion of records flagged as positive that are truly positive.
• PhiUSIIL points: both points are near the upper-right area, showing high precision and high recall for phishing URL detection.
• EMSCAD point: precision is 100.00% and recall is 47.31%; its positive predictions were correct in this test run, but fewer than half of the fraudulent postings were detected.
• ROC-AUC labels: the values printed beside the points provide an additional threshold-independent measure; ROC-AUC is not represented by either axis.

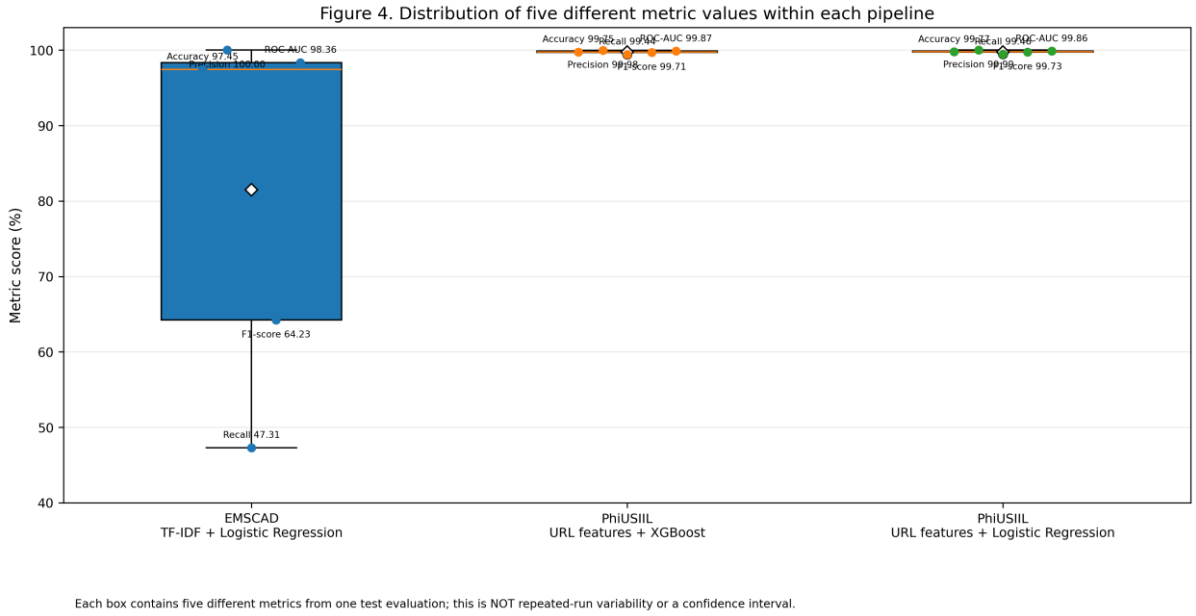


Figure 4. Box plot of the five metric values for each final 70:30 pipeline.

- Box definition: each box summarises five different measures for one pipeline—accuracy, precision, recall, F1-score and ROC-AUC.
- Median: the line inside the box marks the middle value of the five metric values.
- Quartiles and whiskers: the lower and upper edges show the first and third quartiles; the whiskers show the plotted range.
- Mean: the white diamond marks the arithmetic mean of the five metric values.
- Interpretation: EMSCAD has a wider spread because recall and F1-score are lower than its other measures, while the PhiUSIIL values cluster near the top.
- Limitation: this plot compares different metrics from one evaluation; it does not show repeated-run variation, confidence intervals or model stability.

5.1 ROC curve analysis

The ROC curves below are calculated from the actual test labels and per-record positive-class prediction scores exported by the three 70:30 model runs. They are not reconstructed from the AUC summary values. For EMSCAD, positive class 1 means a fraudulent job posting. For both PhiUSIIL pipelines, positive class 1 means a phishing URL after the source label mapping described in Section 2.

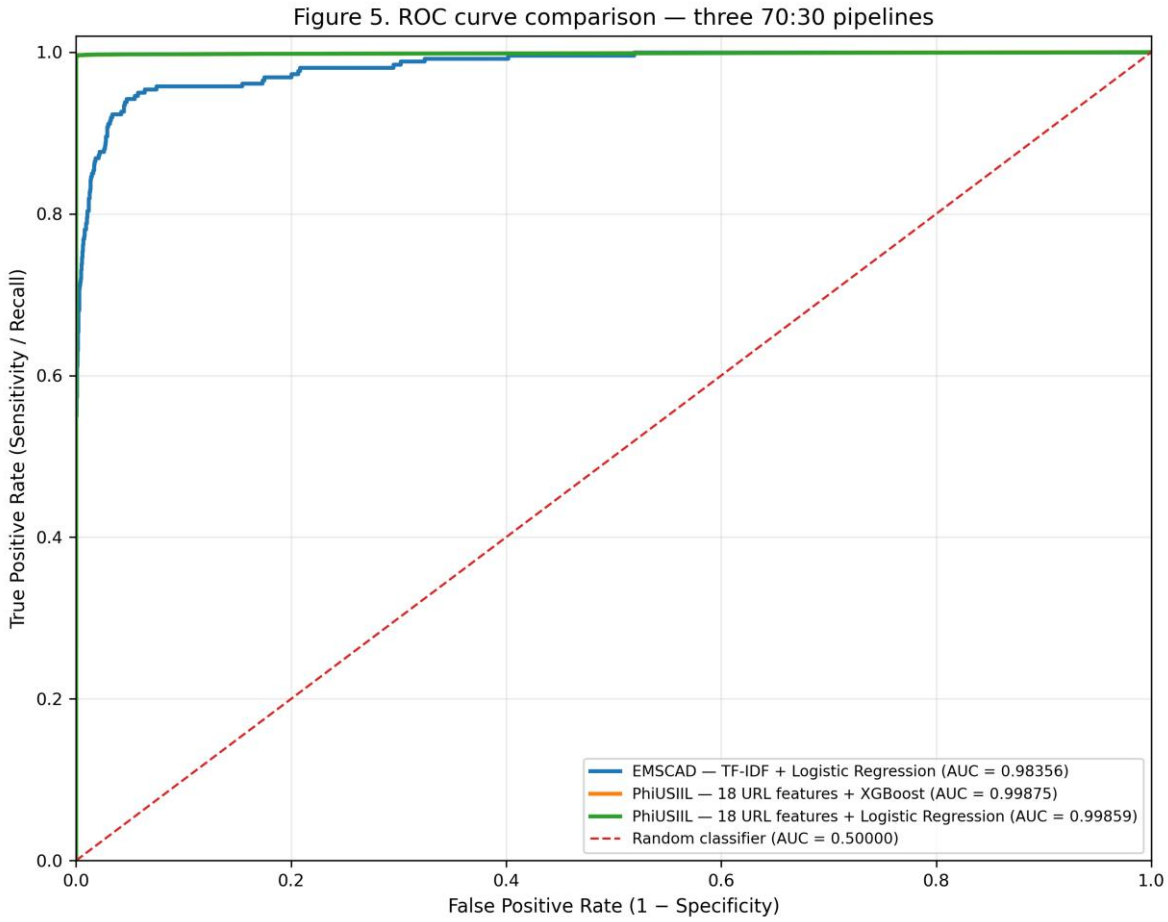


Figure 5. ROC curves of all three final 70:30 pipelines, with measured AUC values in the legend.

- ROC curve definition: the curve plots true positive rate (TPR, equal to recall) against false positive rate (FPR) as the decision threshold changes.
- Diagonal reference: the diagonal line represents random ranking, with an AUC of 0.50. Curves above it indicate better-than-random discrimination.
- PhiUSIIL curves: both are very close to the upper-left corner and have AUC values of approximately 0.999.
- EMSCAD curve: its AUC is approximately 0.984, indicating strong overall ranking performance, although recall at the selected threshold is lower.
- Important distinction: ROC-AUC evaluates ranking across many thresholds, while the recall in the results table is measured at one selected threshold.

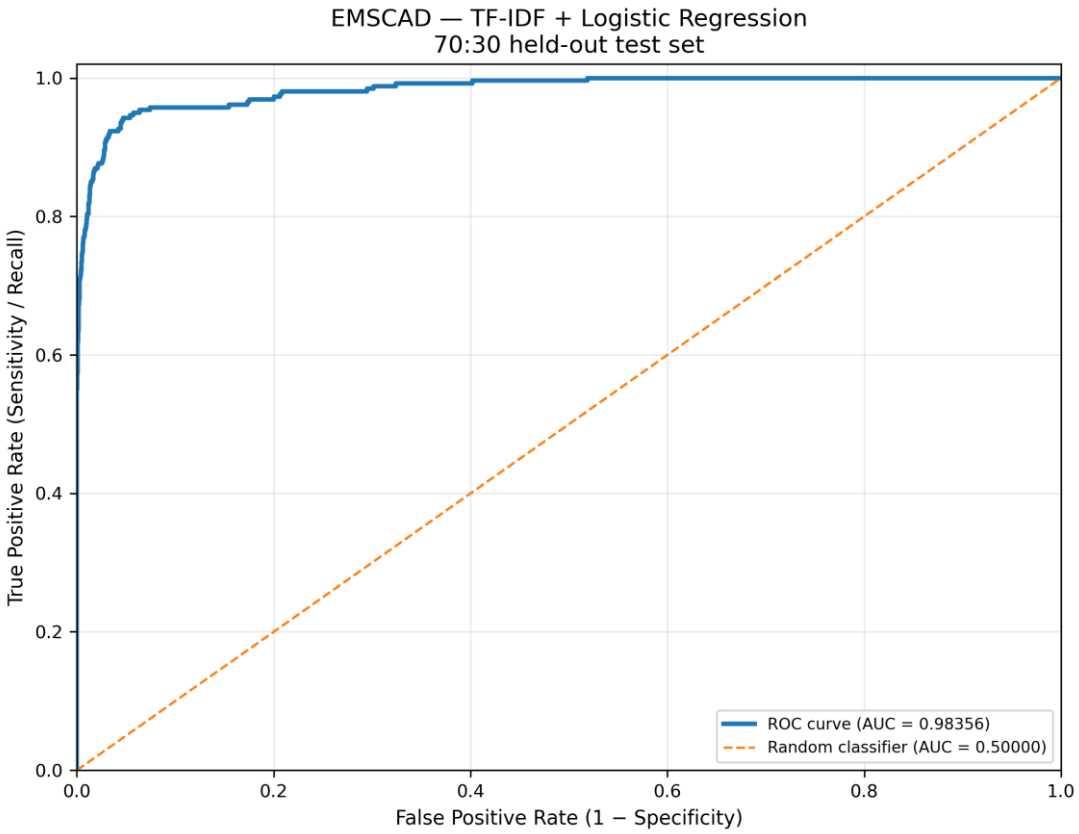


Figure 6. ROC curve for EMSCAD — TF-IDF + Logistic Regression (70:30 held-out test set).

- Dataset and test size: the curve uses 5,364 held-out job postings, including 260 fraudulent and 5,104 legitimate records.
- Axes: the horizontal axis is false positive rate (FPR); the vertical axis is true positive rate (TPR), which is the same as recall.
- Curve meaning: the curve shows the trade-off between detecting fraudulent postings and incorrectly flagging legitimate postings as the threshold changes.
- AUC definition: the area under the ROC curve is 0.98356 (98.356%), showing strong ranking discrimination across thresholds.
- Result interpretation: despite the strong AUC, recall at the selected operating threshold is 47.31%, so the model still misses some fraudulent postings.

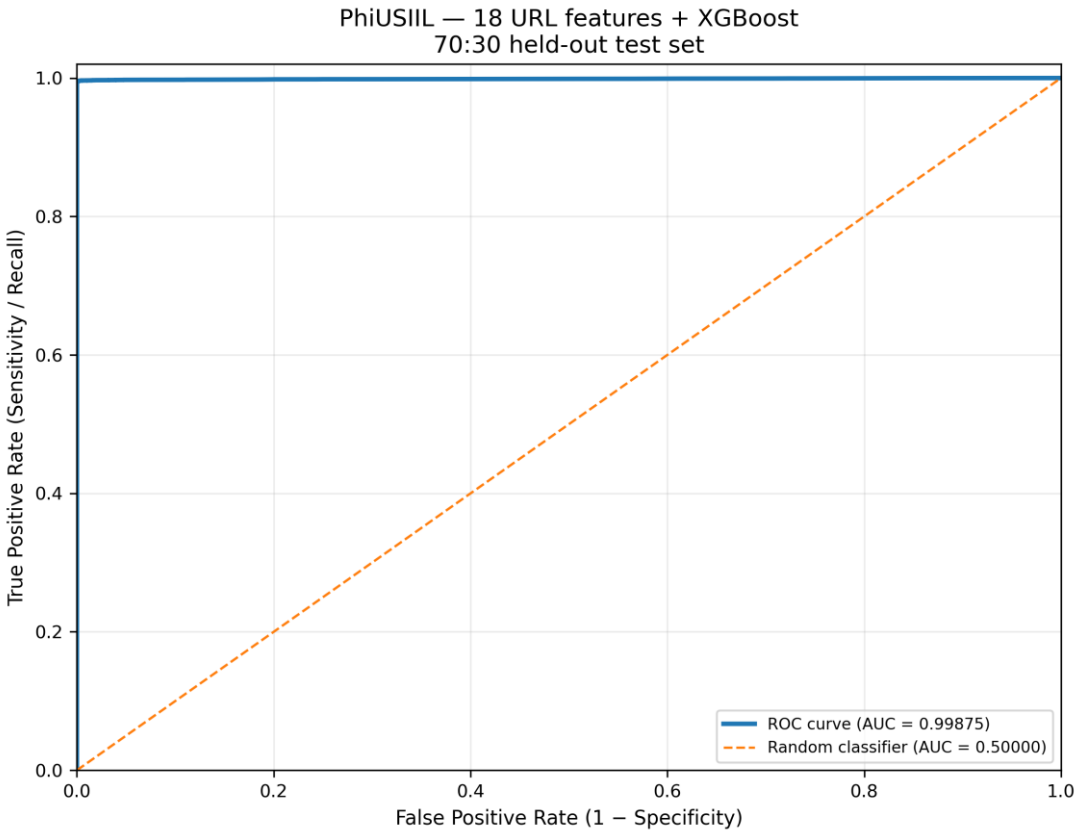


Figure 7. ROC curve for PhiUSIIL — 18 URL features + XGBoost (70:30 held-out test set).

- Dataset and test size: the curve uses 70,739 held-out URLs, including 30,284 phishing URLs and 40,455 legitimate URLs.
- Axes: the horizontal axis is false positive rate (FPR); the vertical axis is true positive rate (TPR), or recall.
- Curve meaning: a curve closer to the upper-left corner indicates that more phishing URLs can be detected while keeping the false-positive rate low.
- AUC definition: the measured area under the curve is 0.99875 (99.875%).
- Result interpretation: this indicates very strong separation of phishing and legitimate URLs within this particular test partition.

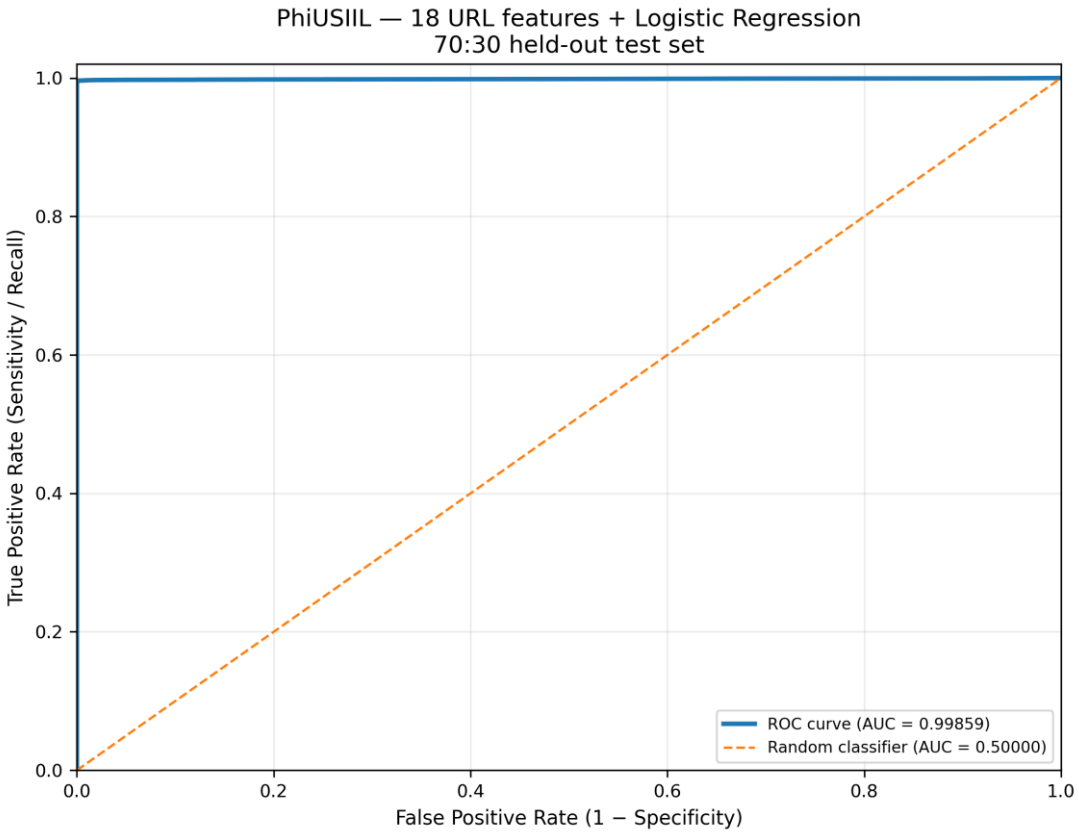


Figure 8. ROC curve for PhiUSIIL — 18 URL features + Logistic Regression (70:30 held-out test set).

- Dataset and test size: the curve uses the same 70,739-record test partition, containing 30,284 phishing URLs and 40,455 legitimate URLs.
- Axes: the horizontal axis is false positive rate (FPR); the vertical axis is true positive rate (TPR), or recall.
- Curve meaning: the closer the curve is to the upper-left corner, the stronger the model's ability to identify phishing URLs while limiting false alarms.
- AUC definition: the measured area under the curve is 0.99859 (99.859%).
- Comparison note: this value is close to the XGBoost result. The small difference does not establish a practically meaningful advantage without repeated or independent evaluation.

6. Discussion

6.1 Discussion of EMSCAD results

The EMSCAD pipeline demonstrates why accuracy alone is not sufficient for evaluating fraud detection. Across the five-split experiment, accuracy remains above 96.9%, and precision is reported as 100.00%, but recall ranges from 36.49% to 47.31%. At the 70:30 split, the confusion matrix contains 137 false negatives and no false positives. The classifier is therefore conservative at the selected threshold: when it flags a posting as fraudulent, the prediction is correct in this test set, but many fraudulent postings are not flagged.

This behaviour is consistent with the challenge of identifying a minority class in recruitment-fraud data. A model can achieve high overall accuracy by correctly classifying the larger legitimate class while missing a meaningful share of fraudulent records. For this reason, recall and F1-score are important measures for the EMSCAD task. Threshold tuning and class weighting were examined in a controlled experiment, with the results reported in Section 7.

The controlled improvement experiment compared word TF-IDF n-grams (1–2), word TF-IDF n-grams (1–3), and character TF-IDF n-grams (3–5), with and without class balancing. The word TF-IDF (1–3) balanced configuration with a validation-selected threshold of 0.57 produced the best F1-score among the reported tuned configurations. Its recall increased substantially compared with the baseline, while precision decreased because more postings were flagged as potentially fraudulent. This is an explicit precision–recall trade-off rather than an improvement in every metric.

6.2 Discussion of PhiUSIIL results

The PhiUSIIL URL-feature pipelines achieve high scores in the reported random stratified test evaluation. In the latest 70:30 run, XGBoost reports 99.75% accuracy, 99.98% precision, 99.44% recall and 99.87% ROC-AUC. Logistic Regression reports 99.77% accuracy, 99.99% precision, 99.46% recall and 99.86% ROC-AUC. These results show that the 18 engineered URL features contain strong predictive information for the labels in this dataset.

The dataset diagnostics identified 235,795 records, no missing values in the selected 18 URL features, 203,176 duplicate engineered feature rows, and 154 unique feature vectors associated with both labels. Identical engineered feature vectors do not necessarily mean that the underlying URLs are identical, because the diagnostic operates on the selected numerical representation rather than raw URL strings. These findings justify an additional grouped evaluation, but do not by themselves prove data leakage.

PhiUSIIL diagnostic	Random stratified 70:30	Grouped by identical feature vectors
Accuracy	99.765%	99.758%
Precision	99.993%	99.990%
Recall	99.459%	99.552%

F1-score	99.725%	99.771%
ROC-AUC	99.859%	99.888%
False positives	2	3
False negatives	164	142

The grouped-split results are close to the random-split results. This is a useful diagnostic showing that grouping identical engineered feature vectors did not materially reduce the reported metrics in this experiment. It is not proof that the model generalises to unseen domains, later time periods or independent URL sources. A future evaluation should use raw URL identifiers, time-based partitions and URLs collected independently of the training dataset.

6.3 Comparison of the two datasets

The EMSCAD and PhiUSIIL scores differ because the models solve different problems using different input representations. EMSCAD relies on natural-language job-advertisement content. Fraudulent postings can use ordinary language, imitate legitimate employers and vary substantially in wording. TF-IDF represents word and n-gram frequency patterns, but it does not automatically capture every contextual or organisational signal that may distinguish a genuine employer from a scammer.

PhiUSIIL uses numerical characteristics extracted from URLs, such as URL length, domain length, subdomain count, obfuscation indicators, character ratios and HTTPS status. The measured dataset may contain highly informative patterns in these features. The high PhiUSIIL scores should therefore be interpreted within that dataset and feature construction, not as evidence that URL detection is universally easier than job-scam detection.

The two pipelines should be viewed as complementary components of JobGuard AI. Their results are not a direct algorithm-versus-algorithm comparison because the records, labels and feature spaces are different.

6.4 Research implications and limitations

The EMSCAD dataset and its recruitment-fraud task are grounded in the work of Vidros et al., who introduced a public collection of annotated online job advertisements. The PhiUSIIL dataset is described by Prasad and Chandra as a large collection of legitimate and phishing URLs with a diverse set of URL and security features. The JobGuard experiments apply selected feature representations and classifiers to these public datasets; their measured results should not be presented as a reproduction of the original papers' full methods.

The principal limitation in the EMSCAD baseline is missed fraudulent postings at the default threshold. The controlled improvement reduces false negatives but increases false positives, so the operating threshold should be selected according to the application's tolerance for missed scams and unnecessary warnings. The PhiUSIIL results require additional validation on independent and newer URLs. Dataset duplication at the engineered-feature level, possible source-specific patterns and changes in attacker behaviour can affect generalisation.

A further limitation is that a high ROC-AUC does not determine the best operational threshold by itself. Threshold selection should use validation data and should be checked against confusion-matrix counts, precision, recall and the consequences of false alarms and missed detections. Results from a single split also do not provide repeated-run uncertainty estimates.

7. Implemented Improvements and Output Evaluation

Four improvement activities were implemented and evaluated in the project experiments. They are retained here as separate changes because they address different parts of the detection pipeline and data-validation process.

No.	Implemented change	What was done
1	Validation-based threshold tuning	Selected the decision threshold using a validation subset of the training data instead of relying only on 0.50.
2	Class-imbalance handling	Used balanced Logistic Regression to give more importance to fraudulent job postings during model fitting.
3	Text feature comparison	Compared word TF-IDF (1–2), word TF-IDF (1–3) and character TF-IDF (3–5) representations.
4	False-negative and data-quality review	Reviewed missed fraudulent postings and false positives, and examined duplicate engineered URL features with grouped evaluation.

7.1 EMSCAD output evaluation

The EMSCAD controlled experiment evaluated the baseline and the selected improved configuration on the same held-out 70:30 test set. Threshold selection was performed using validation data from the training portion, not the test set. The selected configuration was word TF-IDF (1–3) with balanced Logistic Regression and a threshold of 0.57.

Metric	Baseline	Improved configuration
Configuration	Word TF-IDF (1–2), unweighted, threshold 0.50	Word TF-IDF (1–3), balanced, threshold 0.57
Accuracy	97.35%	98.51%
Precision	100.00%	83.33%
Recall	45.38%	86.54%
F1-score	62.43%	84.91%

ROC-AUC	98.46%	98.86%
False negatives	142	35
False positives	0	45

Recall increased by 41.16 percentage points, and the number of missed fraudulent postings decreased from 142 to 35 (107 fewer false negatives). Accuracy and F1-score also increased. Precision decreased from 100.00% to 83.33%, and false positives increased from zero to 45. The improved configuration therefore detects substantially more fraudulent postings at the cost of additional legitimate postings being flagged. This trade-off should be considered when choosing the production threshold.

7.2 PhiUSIIL output evaluation

For PhiUSIIL, the output evaluation focused on label mapping, missing values, repeated engineered feature vectors and the difference between a standard stratified split and a grouped split. The grouped split keeps identical 18-feature vectors together rather than allowing matching feature vectors to appear on both sides of the split.

Evaluation	Accuracy	Precision	Recall	F1-score	ROC-AUC	FP	FN
Random stratified 70:30	99.765%	99.993%	99.459%	99.725%	99.859%	2	164
Grouped by identical feature vectors	99.758%	99.990%	99.552%	99.771%	99.888%	3	142

The grouped evaluation produced results close to the random split. This diagnostic does not establish that every record is independent or that the model will maintain the same scores on future URLs. The available feature-based diagnostic also cannot confirm exact URL duplication because the analysis is based on engineered features.

8. References

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